Yuxin Tang yuxintang-ml.github.io

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EDUCATION

Ph.D.	Computer Science, Rice University, 2018-2024
B.S.	Computer Science, Shanghai Jiao Tong University, 2014-2018

WORK EXPERIENCE

Amazon AGI Foundations, *Applied Scientist*, New York, NY *2025.01–Present*

• Researched novel compute-optimal **fine-grained** MoE model scaling strategy for multimodal pretraining with hardware-aware joint optimization, analyzing the trade-offs between **TPP**, **MoE expert sparsity**, **loss-free load balancing**, **token-choice routing** and hardware features to guide architecture sparsity design.

• Developed hardware-aware **performance model** to predict accurate training MFU across **2,000** B200 GPUs, incorporating comp-comm overlapping, 5D parallelism. Boosting 10%+ MFU by fusing computation into persistent kernels.

Bosch Center for Artificial Intelligence (BCAI), *Research Intern*, Sunnyvale, CA *2024.05–2024.08*

• Work on an algorithmic framework for automatic prompt optimization and prompt tuning.
• Deploy prompt optimization framework to help with Bosch's internal document queries.

Visa Research, *Research Intern*, Palo Alto, CA *2023.05–2023.08*

• Design algorithms for subgraph pattern discovery within trillion-sized transaction graphs.
• Implement biclique computation framework designed to efficiently handle bipartite graphs.

SELECTED PUBLICATION

JMLR'26 **Compute-Optimal Is Not Cluster-Optimal: Systems-Aware Scaling for Sparse Mixture-of-Experts.** [link]
Soumajyoti Sarkar, Yuxin Tang, Sheng Zha.

Blog'26 **Introducing Harbor-Index: A Compact, Diverse, Challenging, and High-Quality Benchmark for Agentic Evaluation.** [link]
Harbor Team.

VLDB'26 **Automated Tensor-Relational Decomposition for Large-Scale Sparse Tensor Computation.** [link]
Yuxin Tang, Zhiyuan Xin, Zhimin Ding, Xinyu Yao, Daniel Bourgeois, Tirthak Patel, Chris Jermaine.

ICLR'26 **DOPPLER: Dual-Policy Learning for Device Assignment in Asynchronous Dataflow Graphs.** [link]
Xinyu Yao, Daniel Bourgeois, Abhinav Jain, Yuxin Tang, Jiawen Yao, Zhimin Ding, Arlei Silva, Chris Jermaine.

VLDB'25 **EinDecomp: Decomposition of Declaratively-Specified Machine Learning and Numerical Computations for Parallel Execution.** [link]
Daniel Bourgeois, Zhimin Ding, Dimitrije Jankov, Jiehui Li, Mahmoud Sleem, Yuxin Tang, Jiawen Yao, Xinyu Yao, Chris Jermaine.

BigData'24 **Monarch: Distributed Butterfly Counting for Large-scale Bipartite Graph.** [link]
Yuxin Tang, Mangesh Bendre, Mahashweta Das.

ICML'24 **Soft Prompt Recovers Compressed LLMs, Transferably.** [link]
Zhaozhuo Xu*, Zirui Liu*, Beidi Chen, Shaochen Zhong, Yuxin Tang, Jue Wang, Kaixiong Zhou, Xia Hu, Anshumali Shrivastava.

ICCV'23 **Federated Learning Over Images: Vertical Decompositions and Pre-Trained Backbones Are Difficult to Beat.** [link]
Yuxin Tang*, Ed Hu*, Anastasios Kyrillidis, Chris Jermaine.

ICML'23 **Auto-Differentiation of Relational Computations for Very Large Scale Machine Learning.** [link]
Yuxin Tang, Zhimin Ding, Dimitrije Jankov, Binhang Yuan, Daniel Bourgeois, Chris Jermaine.

VLDB'22 **Distributed learning of fully connected neural networks using independent subnet training.** [link]
Binhang Yuan, Cameron R. Wolfe, Chen Dun, Yuxin Tang, Anastasios Kyrillidis, Chris Jermaine.

VLDB'21 **Tensor Relational Algebra for Machine Learning System Design.** [link]
Binhang Yuan, Dimitrije Jankov, Jia Zou, Yuxin Tang, Daniel Bourgeois, and Chris Jermaine.

USENIX Sec'20 **Programmable In-Network Security for Context-aware BYOD Policies.** [link]
Qiao Kang, Lei Xue, Adam Morrison, Yuxin Tang, Ang Chen, Xiapu Luo.

OSDI'18 **A Programmable, Hardware-Assisted Network Protocol Fuzzer.** *(Poster)*
Yuxin Tang, Ang Chen.

WUWNet'17 **Exploring Simulation of Software-Defined Underwater Wireless Networks.** [link]
Li Wei, Yuxin Tang, Yuching Cao, Zhaohui Wang, Mario Gerla.

SERVICE

Conference Reviewer:
ICLR 2021–2026, ICML 2020–2026, NeurIPS 2021–2026

Session Chair:
VLDB 2023, MLSys 2025

Program Committee:
CGO 2023, UDM-AAAI 2023, PLDI 2023

REFERENCES

Professor Chris Jermaine, Department of Computer Science, Rice University
Professor Xia "Ben" Hu, Department of Computer Science, Rice University
Professor Arlei Silva, Department of Computer Science, Rice University
Professor Beidi Chen, Department of Electrical and Computer Engineering, Carnegie Mellon University (CMU)
Professor Binhang Yuan, Department of Computer Science & Engineering, Hong Kong University of Science and Technology (HKUST)